IETI College of Science and Technology

#34 Lark St., Sta. Elena, Marikina City | Tel No. 8682-2984

CookLabs: A Web-Based Learning Platform

for Cookery NC II students of IETI Marikina

A Thesis presented to the faculty of

IETI College of Science and Technology

Marikina Campus

In Partial Fulfillment of the Requirements for the degree of

Bachelor of Science in Information Technology

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RECOMMENDATION FOR ORAL EXAMINATION

This Thesis paper entitled **COOKLABS: A WEB-BASED LEARNING PLATFORM FOR COOKERY NC II STUDENTS OF IETI MARIKINA**, prepared and submitted by **RENZ AARON M. MENDIOLA**, in partial fulfillment of the requirements of Bachelor of Science in Information Technology, has been examined and is hereby recommended for oral examination.

RORY TAPANG, MPA
Thesis Adviser / Professor

APPROVAL SHEET

The Thesis entitled **CookLabs: A Web-Based Learning Platform for Cookery NC II students of IETI Marikina**, prepared and submitted by **Renz Aaron M. Mendiola** has been approved by the Committee on Oral Examination with a grade of _____ on 10, May 2025.

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

Introduction

Cooking is a skill that requires more than just following recipes and it takes practice, clear instructions, and the right knowledge to do it well. For Cookery NC II students, having access to helpful learning tools is important, especially when they need to study both the theory and the hands-on parts of cooking. But sometimes, it can be hard to find materials that are easy to understand or available anytime they need them.

To help with this, CookLabs was created, a web-based learning platform made for Cookery NC II students at IETI Marikina. The goal of this platform is to make learning more flexible and accessible. It includes modules and quizzes based on the standards of TESDA. With CookLabs, students can study at their own pace and go back to lessons whenever they need to.

The purpose of this study is to determine how effective CookLabs is in instructing cooking to students. It also examines how learning can be enhanced by digital assets such as this one, especially in courses that require practical skills.

Background of the study

Cookery NC II is a technical-vocational course offered by TESDA that teaches students the basic skills of cooking and food preparation. This course is essential for people who wish to learn useful skills for daily living as well as those who wish to work in the food and hospitality industry. In IETI Marikina, many students take this course with the hope of gaining experience and learn useful techniques they can use in the real world.

However, learning is not always easy, especially when students don't have enough access to materials or time for practice outside the classroom. Traditional methods, such as printed modules or classroom lectures, may not always meet the needs of every student especially in situations where face-to-face classes are limited or when students need extra support to study at their own pace. Because of this, there is a need for a learning tool that can support students anytime and anywhere.

Today, many schools and students are using technology to help with learning. With internet access, students can now study online through websites and apps. That's why CookLabs, a web-based learning platform, was created to help Cookery NC II students study their lessons more easily. It includes features and lessons that students can access anytime, even outside the classroom.

This study was made to find out if CookLabs can help students learn better and make their study experience easier and more enjoyable.

Theoretical Framework

The study is based on several important learning theories that explain how students learn best. These theories guide the design of the CookLabs: A Web-Based Learning Platform for Cookery NC II students of IETI Marikina.

Constructivist Theory of Learning (Jerome Bruner, Lev Vygotsky, and Jean Piaget). The basic idea of this theory is the concept that students actively create their own knowledge through experience and self-study instead of just absorbing it. This theory believes that learning is a personal process in which people connect new ideas to what they already know, usually through experimentation and practical experience.

This theory emphasizes how important it is for students to take an interest in their education by practicing cooking techniques on their own in out-of-school setting. Students interact with the learning materials by experimenting with recipes and techniques independently, rather than depending only on guided lessons or direct instruction. Through exploration, error, and learning from experiences, this self-practice enables students to develop their knowledge and abilities over time.

Technology Acceptance Model (TAM) (Fred Davis). The Technology Acceptance Model explains how users come to accept and use new technology. It highlights two important factors:

- **Perceived Usefulness** – The degree to which a person believes that using a technology will help improve their performance.

- **Perceived Ease of Use** – How easy and user-friendly the person finds the technology.

This study examines whether CookLabs is perceived as useful and easy to use by the students, which influences their willingness to adopt it as a learning tool.

By combining these two theories, the study aims to understand how CookLabs supports self-directed learning and how likely students are to accept and use the platform in their cookery education.

Conceptual Framework

The Input – Process – Output (IPO) System was used in describing the Conceptual Framework of the study. The INPUT consists of Demographic Profile of respondents and Assessment in terms of accessibility of learning materials, progress tracking, and assessments tools delivered by CookLabs. The PROCESS involves the development and implementation of CookLabs: A Web-Based Learning Platform for Cookery NC II students of IETI Marikina, which includes key features such as learning materials management, progress tracking, and assessment tools. The OUTPUT is the proposed enhancements in the learning platform to better support Cookery NC II students.

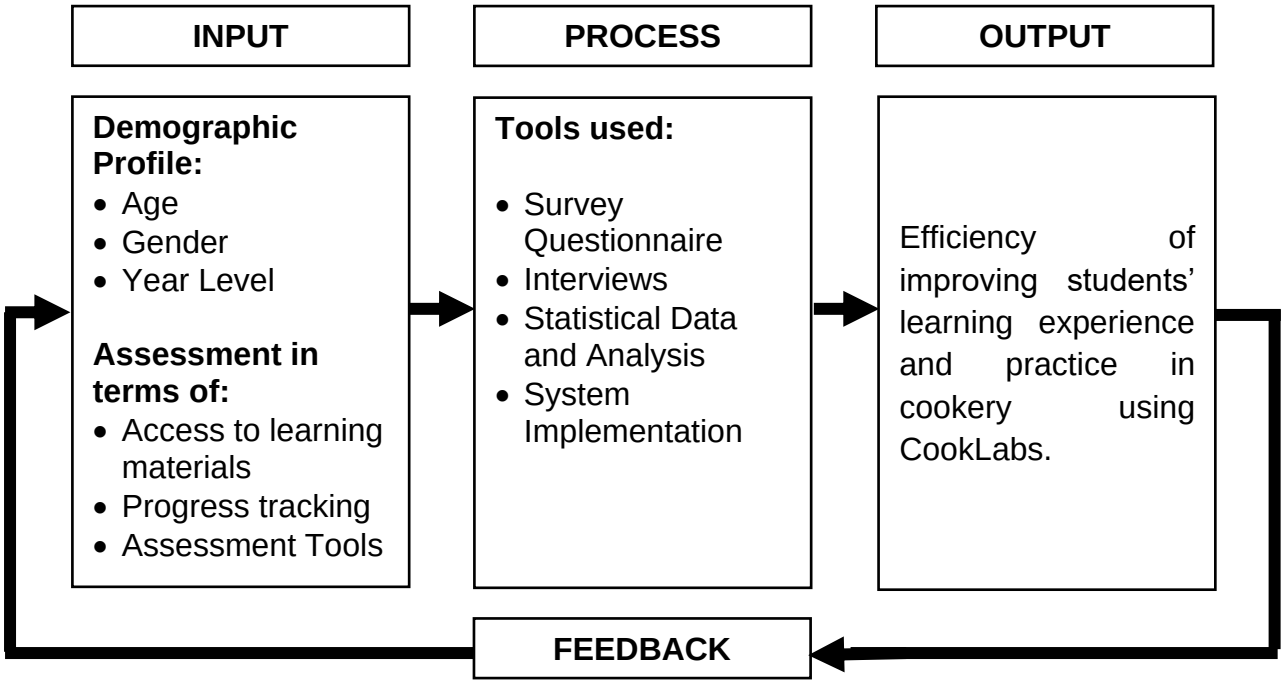

Figure 1: Paradigm of the Study

Statement of the Problem

Cookery NC II is a hands-on course that teaches students with the fundamental knowledge and skills to become competent cooks, chefs, or food preparers. But for many students, learning can be challenging especially when they don't have time in class or enough access to materials to practice outside the classroom. These challenges can affect how well they understand and apply what they're supposed to learn.

Nowadays, students often turn to technology to help them study. Because of this, there's a need for a simple and accessible learning platform

that allows them to practice on their own and study whenever it's convenient for them.

This study focuses on developing and evaluating CookLabs, a web-based learning platform made to help Cookery NC II students at IETI Marikina. The goal is to support their learning by giving them resources they can access anytime. Specifically, this study seeks to answer the following questions:

1. What is the demographic profile of the respondents in this study in terms of:
   1.1 Age;
   1.2 Gender; and
   1.3 Year Level?
2. How do the respondents assess the efficiency of homeschooling in terms of the following:
   2.1 Access to learning materials;
   2.2 Progress tracking; and
   2.3 Assessment tools?
3. Are there any significant benefits in using CookLabs: A Web-Based Learning Platform for Cookery NC II students of IETI Marikina in terms of accessibility of learning materials, progress tracking, and assessment tools?

Hypothesis

There are no significant benefits in using CookLabs: A Web-Based Learning Platform for Cookery NC II students of IETI Marikina in terms of accessibility of learning materials, progress tracking, and assessment tools.

Significance of the Study

This study is important because it offers a way to support Cookery NC II students in learning and practicing on their own. With limited time and resources during classes, students often need extra support at home. CookLabs provides a flexible and easy-to-use platform where they can access lessons, review recipes, and monitor their progress at their own pace. It is designed to support independent learning and help students feel more confident in building their cooking skills.

The results of this study will benefit the following:

Cookery NC II Students. The main goal of this platform is to help students improve their learning experience. With CookLabs, they can go back to lessons anytime, practice cooking steps on their own, and feel more prepared in their practical activities. It can also help them stay on track even when they're not in the kitchen or attending face-to-face classes

Instructors. Teachers can use the platform to give extra materials to their students and monitor what topics students are focusing on. Although it is meant for self-practice, the platform can help instructors identify areas where students may need more support. It also serves as a tool to recommend when students need additional practice or review.

The Researcher. Through this project, the researcher is able to apply their skills in designing and building a web-based system. It allows the researcher to explore how technology can help improve education and respond to student learning challenges. Working on a project that has practical applications also helps the researcher grow personally and academically.

Future Researchers. This study can serve as a useful reference for others who want to develop similar platforms for technical-vocational education. It can guide them in understanding what features are helpful, what challenges may arise during system development, and how to assess if a learning tool is effective.

In summary, this study is significant because it helps improve the way cookery students learn, gives teachers a helpful support tool, and shows how simple technology can be used to improve learning in vocational education.

Scope and Limitation

CookLabs is a web-based learning platform designed primarily for Cookery NC II students of IETI Marikina. The system will have features including profile management, module management, assessment tools, and progress tracking. Though this system will mainly be used by students, teachers and/or instructors may as well benefit from this through features for assessment and progress tracking. This study respondents were limited depending on the number of students that are enrolled in Cookery NC II program at IETI Marikina. Respondents consist of male and female participants, specifically students enrolled in Cookery NC II program in IETI Marikina.

However, this study has some limitations including that the system will not provide an offline mode as it is purely web-based. The platform assessment and evaluation is limited to multiple choice quiz and does not have any other form of assessment. Lastly, the system was tested only by a limited number of students from one school, so the findings may not apply to all cookery students in general. Despite these limitations, the platform aims to offer a helpful learning tool that can support students in their study and practice.

Definition of Terms

For the purpose of clarity and understanding, the following terms were given their conceptual and operational definitions:

Accessibility – Refers to how easily students can access and use the platform's learning materials anytime using an internet-connected device.

Assessment Tools – Features in CookLabs such as quizzes or exercises that help evaluate student learning and progress.

Blended Learning – A combination of online digital learning and traditional classroom methods, allowing students more flexibility in their studies.

Cookery NC II – A TESDA-accredited course that teaches students essential knowledge and skills in food preparation and cooking techniques.

CookLabs – A web-based learning platform created for this study to help Cookery NC II students learn through online lessons, recipe guides, and self-practice tools.

Content Management – A system feature that allows teachers or administrators to upload, organize, and update learning materials like lessons and recipes.

Digital Learning – A learning method that uses digital tools, apps, or platforms, like CookLabs, to deliver educational content over the internet.

Ease of Use – How simple and user-friendly the system is for students to navigate, which affects their willingness to use it regularly.

Independent Learning – A way of studying where students take control of their own learning pace and time without direct teacher supervision.

Learning Materials – Digital content used for study, such as lessons, recipes, videos, or review guides made available through the platform.

Learning Platform – An online system that delivers lessons, assessments, and other learning tools to support students outside the traditional classroom.

Monitoring – The process of tracking student activity or progress to identify which lessons have been completed and what areas need improvement.

Practice Modules – Interactive activities or step-by-step exercises that students can follow on their own to practice cooking techniques.

Progress Tracking – A feature of the platform that records and displays the student's learning progress, completed lessons, and performance in assessments.

Self-Practice – The ability of students to study and try cooking steps independently, using platform materials, without needing direct supervision.

TESDA – Technical Education and Skills Development Authority, the agency in charge of regulating and promoting technical-vocational education in the Philippines.

Web-Based System – A system that operates online through a browser without the need for installation, making it accessible anytime with an internet connection.

CHAPTER II

Review of Related Literature and Studies

Related Literature

Foreign

Mulyatiningsih et al. (2021) investigated the characteristics of online learning that students found enjoyable, as a result of vocational students transitioning to the online modality, due to the COVID-19 pandemic. It was found that students found Moodle to be enjoyable for structured learning, Google Meet to be enjoyable for ease of access, and WhatsApp to be enjoyable for daily interaction. Furthermore, the research said that having short content pieces and a blend of online learning with both synchronous and asynchronous ways of learning increased student involvement.

Grasielda and Yuwono (2022) studied the development of learning strategies to incorporate in online culinary practice classes utilizing Massive Open Online Courses (MOOC). They argued, based on their findings, that an appropriately designed MOOC can meet expectations of stakeholders involved in conducting the practicum class, while still being able to conduct the practicum class effectively, and with limited time compared to a conventional practicum class. The research also pointed out that for the implementation to be a

success, planning holistically was necessary, and communication once students begin the MOOC had to be done regularly with the lecturer.

Asenahabi et al. (2022) conducted a literature review examining the usability of Learning Management Systems (LMS) in supporting online learning within higher education. Their work highlighted that technological advancements—accelerated by the COVID-19 pandemic—have significantly influenced the adoption of technology-enhanced learning in universities. The authors found that inadequate consideration of LMS usability during implementation may negatively impact the learning experience, leading lecturers to abandon institutional systems in favor of more user-friendly alternatives. The review also emphasized that although top global institutions increasingly offer online courses and programs, many lecturers still do not use LMS platforms to their full potential. According to the researchers, understanding LMS usability attributes is essential for universities seeking to enhance online learning environments and ensure successful system adoption.

Local

Palomeras (2024) created and assessed a digitalized learning module on Cookery for Grade 11 students in Antipolo City. The module was rated very satisfactorily by teachers and experts, a reflection of its efficacy in enhancing students' academic performance and critical thinking skills. The study

concluded that digitalized learning modules could enhance academic performance, even provided the learning is taking place online.

Egano (2024) evaluated the modular approach for Grade 7 TLE Cookery instructions in Cebu City. The study reported that indicators (and students) experienced challenges using the modular approach as an adaptation of on-site lessons in face-to-face classes, which affected students' academic performance. It recommended some contextualized practical learning activities to improve learning ability.

Roa et al. (2020) examined the effectiveness of the Learning Management System used at the University of Science and Technology of Southern Philippines in the Cagayan de Oro and Villanueva campuses. Their findings showed that LMS effectiveness depends heavily on user-friendly interfaces, reliable technical support, and active faculty involvement in facilitating learning. The study also highlighted ongoing challenges in digital infrastructure and teacher training that affect LMS implementation.

Related Studies

Foreign

Daryanto, Siregar and Januariyansah (2022) performed a meta-analysis to assess how e-learning influenced Vocational Education and Training. The meta-analysis consisted of 18 studies, with the results showing that e-learning

has a practice impact on the theoretical learning outcomes, and e-learning also positively affected students' motivation and interests. On the other hand, e-learning is less, effective for traditional learning practicum component, whereby they suggested instructor led digital simulations, and web-based programming to support learnings for practicum based on practical components of learning.

Saripudin et al. (2020) surveyed vocational school teachers in West Java which sought out the teachers' perceptions towards e-learning during the COVID-19 pandemic. Although the survey concluded that majority of teachers considered that they could manage e-learning implementation, as much as 73% of teachers indicated that the content was difficult to be comprehended by students - indicating that there was a need that teachers were well prepared and supported to be able to effectively deliver e-learning content to learners.

Zhang et al. (2024) sought to understand the effect digital technology usage has on higher vocational student satisfaction with learning by examining whether learning experience and learning engagement mediated the effect. Using a structural model analysis, their results indicated that digital technology usage has a positive influence on the learning experience and engagement, which both satisfied students learning. Thus, their study demonstrated that there are real benefits of integrating various digital technologies in order to develop better learning experience within vocational education.

Local

Tus et al. (2021) examined the relationship between Learning Management System usage and the academic performance of 188 Filipino senior high school students. The researchers found, using the Pearson correlation coefficient, that there was no significant relationship between LMS usage and students' academic performance. Their findings indicated that LMS activity alone may not be a strong predictor of academic outcomes. The study also suggested that other factors such as motivation, self-efficacy, and learner control may play a more substantial role in students' academic success.

Corriente, Englis, and Pinatil (2022) examined the challenges of cookery teachers in a state-owned basic education institution in the Philippines during the COVID-19 pandemic. The researchers found that the teachers faced challenges that included limited access to digital/online tools, the difficulties in preparing and retrieving the modules, and purchasing equipment and materials, as well as the need for concrete strategies to ensure learning for students. The researchers stressed the need for adequate provision of resources and support for the teachers in order to determine the extent or goals of cookery education in a challenging context.

Laspiñas and Almerez (2024) examine users' experiences with a Learning Management System (LMS) in a higher education institution in Cebu, Philippines. Their study focuses on key factors influencing LMS interaction,

including usability, accessibility, pedagogical integration, and the availability of support and training. While past research highlights issues such as poor internet connectivity and the need for user-friendly interfaces, little is known about how these factors specifically affect Cebu-based institutions. Their work addresses this gap by identifying local challenges and providing insights to improve LMS utilization and user satisfaction.

Synthesis

The reviewed literature and studies reveal a growing integration of digital tools, Learning Management Systems (LMS), and online modalities in education, particularly in vocational and technical subjects such as cookery. Foreign studies, such as those by Mulyatiningsih et al. (2021) and Grasielda and Yuwono (2022), emphasize that the effective use of platforms like Moodle, Google Meet, and MOOCs enhances student engagement, learning flexibility, and the ability to conduct practicum-based classes remotely. Similarly, Asenahabi et al. (2022) highlighted the importance of LMS usability, noting that poorly designed systems can reduce adoption and affect learning outcomes. Local literature, including studies by Palomeras (2024) and Roa et al. (2020), supports these findings by demonstrating that digitalized modules and user-friendly LMS interfaces improve academic performance and facilitate learning, though challenges such as limited teacher training and digital infrastructure persist.

Meanwhile, related studies suggest that the effectiveness of LMS and digital learning tools is not solely determined by their usage. Tus et al. (2021) found no direct correlation between LMS usage and student academic performance, indicating that factors such as motivation, self-efficacy, and learner control significantly influence outcomes. This aligns with foreign studies (Daryanto et al., 2022; Zhang et al., 2024) showing that digital technology positively impacts student engagement and learning satisfaction, especially when combined with supportive teaching practices. Additionally, research on practical subjects, such as cookery, underscores the necessity of contextualized and interactive learning materials. Studies by Corriente et al. (2022) and Laspiñas and Almerez (2024) examine users' experiences with a Learning Management System (LMS) in an institution showing that factors like ease of use, accessibility, teaching integration, and available support shape their overall satisfaction and highlight areas for improvement.

In summary, the synthesis of these literature and studies indicates that while LMS and digital learning tools offer significant opportunities to enhance learning engagement and flexibility, their effectiveness is highly dependent on usability, teacher support, and complementary factors such as student motivation and practical, interactive content. Successful integration of online tools in both local and foreign contexts requires careful planning, regular communication, and continuous support to maximize learning outcomes.

CHAPTER III

METHODOLOGY

This chapter deals with the methods of research used whether it may be historical, descriptive, and experimental or a case study. The techniques used under descriptive research method as well as the data gathering tools and analytic tools used will be further explained in this chapter.

Research Design

This study used a descriptive developmental research design. The goal was to design, build, and evaluate a web-based learning platform that can help Cookery NC II students with their studies. The descriptive part of the study focused on gathering feedback from students about how useful and easy the platform was to use. The developmental part focused on creating CookLabs itself, from planning its features to testing how well it worked.

By combining both, the research was able to look at how the platform met the needs of the students and what improvements could be made. It also helped in understanding whether students were able to learn better when given access to CookLabs outside the classroom.

Respondents of the Study

The respondents of the study included parents and tutors actively involved in homeschooling practices. The participants were selected based on their familiarity or engagement with the application or homeschooling systems in general. The survey was administered through online forms or printed copies, depending on accessibility.

A purposive sampling technique was used to identify individuals who are knowledgeable and capable of providing meaningful feedback about the features and usability of the Modular Homeschool Application. This sampling method ensured that only respondents who directly relate to the study's objectives were included.

Research Instrument

The main research instrument used in this study is a survey questionnaire developed specifically for students enrolled in Cookery NC II program. The questionnaire was carefully designed to evaluate key features of the Cooklabs: A Web-based Learning Platform, including access to learning materials, progress tracking, and assessment tools functionality.

The survey consists of multiple parts, each focusing on a specific feature of the system. Responses are captured using a 5-point Likert scale, ranging from Strongly Disagree to Strongly Agree, allowing respondents to express the level of their agreement or satisfaction with each feature. The instrument was

reviewed by academic advisers and piloted with a small group of users to ensure clarity, relevance, and reliability.

Data Gathering Procedure

Before collecting any data, permission was first requested from the administration of IETI Marikina to conduct the study among Cookery NC II students. Once approved, the researcher introduced the CookLabs web-based learning platform to the students and gave them time to explore and use its features.

After they had time to explore and use the platform, printed survey questionnaires were personally distributed to the students. The researcher explained the purpose of the survey and assured them that their responses would remain confidential and be used only for the study.

All responses were collected and reviewed. The data gathered from the questionnaires were used to analyze how effective and helpful CookLabs was in supporting the students' learning experience.

Statistical Treatment and Analysis

To have a general view of the whole scenario of the study, statistical tool is used. This also includes Likert scale, which will be used by the researcher as a technique to monitor the respondents' interpretation of facts.

Below are the formulas that are to be used to arrive to the computation used by the statistical tool:

1. Percentile Distribution

$$P = \frac{n}{T}(100)$$

Where:

P = Percentage

N = Number of respondents' response

T = Total number of respondents

100 = Constant number

2. Weighted Mean

$$WM = \frac{X1(f)+X2(f)+X3(f)+X4(f)+X5(f)}{N}$$

Where:

N = Total number of respondents

$X_{(\ )}$ = Weight

f = Frequency

3. Likert Scale

With 1 for the lowest and 5 for the highest. The following describes the weight and the mean range of the degree of influence, represented by each point.

LIKERT SCALE		
WEIGHT	**MEAN RANGE**	**DESCRIPTION**
5	**4.50 – 5.00**	**Very Effective**
4	**3.50 – 4.49**	**Effective**
3	**2.50 – 3.49**	**Moderately Effective**
2	**1.50 – 2.49**	**Ineffective**
1	**1.0- 1.49**	**Very Ineffective**

*The term moderately efficient differs from slightly efficient as the slightly efficient is already near the inefficient category, But not totally inefficient.

4. **Independent t-test.** Also called "unpaired two sample", is used to compare the means of two unrelated groups of samples.

$$t = \frac{m_A - m_B}{\sqrt{\frac{S^2}{n_A} + \frac{S^2}{n_B}}}$$

S^2 is an estimator of the common variance of the two samples. It can be calculated as follow:

$$S^2 = \frac{\Sigma(x - m_A)^2 + \Sigma(x - m_B)^2}{n_A + n_B - 2}$$

REFERENCES

Asenahabi, B. M., Ikoha, P. A., & Nambiro, A. W. (2022). Learning management system usability towards online learning: A literature review. School of Science, Technology and Engineering, Alupe University College; School of Computing and Informatics, Kibabii University.

Corriente et al. (2022). *The teaching of cookery course in a state-owned basic education institution in the Philippines during the COVID-19 pandemic*. International Journal of Science and Management Studies, 5(3), 27–37. https://doi.org/10.51386/25815946/ijsms-v5i3p104

Daryanto et al. (2022). *A meta-analysis of the e-learning influence on Vocational Education and Training (VET): Preliminary study of virtual to actualization*. International Journal of Education in Mathematics, Science, and Technology (IJEMST), 10(3), 710–721. https://doi.org/10.46328/ijemst.2509

Egano (2024). *Modular approach in teaching Grade 7 TLE cookery instruction*. International Journal of Innovative Science and Applied Engineering, 2(1), 45–52. https://ijisae.org/index.php/IJISAE/article/view/6921

Grasielda & Yuwono (2022). *Culinary practical online class: A development of study method amid COVID-19 pandemic*. TRJ Tourism Research Journal, 6(1), 107–126. https://doi.org/10.30647/trj.v6i1.137

Laspiñas, R. N., & Almerez, A. U. (2024). Users' experience study on the Learning Management System of a higher education institution in Cebu, Philippines. Psychology and Education: A Multidisciplinary Journal, 17(10), 1062–1066. https://doi.org/10.5281/zenodo.10827162

Mulyatiningsih et al. (2021). *The enjoyable online learning model for vocational students during COVID-19 pandemic*. International Journal of Evaluation and Research in Education, 12(1), 1–10. https://doi.org/10.11591/ijere.v12i1.23122

Palomeras (2024). *Digitalized learning module in cookery for Grade 11 students*. International Journal of Multidisciplinary: Applied Business and Education Research, 5(9), 3554–3560. https://doi.org/10.11594/ijmaber.05.09.14

Roa, M. F. V., Gimeno, E. L. A., Tenorio, C. B., & Malawani, A. D. (2020). Effectiveness of Learning Management System in University of Science and Technology of Southern Philippines Cagayan de Oro and Villanueva Campuses: A policy recommendation. University of Science and Technology of Southern Philippines; Bukidnon State University; Mindanao State University – Iligan Institute of Technology.

Saripudin et al. (2020). *Vocational school teachers' perceptions of e-learning during COVID-19*. Journal of Engineering Education Transformations, 34(Special Issue), 7–13. https://doi.org/10.16920/jeet/2020/v34i0/157844

Tus, J., Cruz, M. A., Espiritu, N. A., & Paras, N. E. (2021). Amidst the online learning modality: The usage of learning management system (LMS) and its relationship to the academic performance of Filipino students. International Journal of Psychology and Behavioral Sciences, 11(4), 9–21.

Zhang et al. (2024). *The effect of digital technology usage on higher vocational student satisfaction: The mediating role of learning experience and learning engagement*. Frontiers in Education, 9, 1508119. https://doi.org/10.3389/feduc.2024.1508119

APPENDIX A

May 16, 2025

IETI College of Science and Technology, Inc.
#34 Lark St., Sta. Elena, Marikina City

Greetings of Peace and Prosperity!

Dear Ma'am/Sir,

I am **Renz Aaron Mendiola,** presently studying as a Third Year BSIT Student in IETI College of Science and Technology at Marikina Campus. In partial fulfillment of my undergraduate studies, I am required to document and construct a computer system related to my thesis research **"CookLabs: A Web-Based Learning Platform for Cookery NC II students of IETI Marikina".**

I kindly request permission to proceed with the project CookLabs: A Web-Based Learning Platform for Cookery NC II students of IETI Marikina and to document the steps involved in building the system, which will be an important in system research in **CAP 1** (CAPSTONE 1) at IETI, #34 Lark St., Sta. Elena, Marikina City.

Sincerely Yours,

_____

Renz Aaron Mendiola
BSIT Student
IETI College of Science and Technology

Noted by:

_____

Rory Tapang, MPA
Thesis adviser

Approved by:

_____

Mrs. Analyn A. Navarrete
School Directress

APPENDIX A

May 16, 2025

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#34 Lark St., Sta. Elena, Marikina City

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Sincerely Yours,

_____

Renz Aaron Mendiola
BSIT Student
IETI College of Science and Technology

Noted by:

_____

Rellie Monroe Navarrete,
MMHoA, MIT, CCNA
Thesis adviser

Approved by:

_____

Mrs. Analyn A. Navarrete
School Directress

APPENDIX B

Good day!

Magandang Umaga!

I am Renz Aaron Mendiola, currently taking up BSIT at IETI College of Science and Technology, Marikina. I am conducting a survey that will assess the efficiency and effectiveness of CookLabs: A Web-Based Learning Platform for Cookery NC II students of IETI Marikina.

Ako po si Renz Aaron Mendiola, kasalukuyang kumukuha ng kursong BSIT sa IETI College of Science and Technology, Marikina. Nagsasagawa po ako ng isang survey sa kahusayan at kung gaano ka-epektibo ang CookLabs: A Web-Based Learning Platform for Cookery NC II students of IETI Marikina.

I will appreciate if you could spare some of your time to answer the questions and complete the table given. Any information obtained in connection with this study will remain private and confidential.

Lubos kong ikagagalak kung makapaglalaan ka ng oras upang sagutan ang mga katanungan at kumpletuhin ang ibinigay na talahanayan. Anumang impormasyon na nakuha sa pag-aaral na ito ay mananatiling probado at kumpidensyal.

Thank you!

Maraming salamat!

Respectfully yours,

Lubos na gumagalang

_____

Renz Aaron Mendiola

BSIT Student – Researcher

APPENDIX C

Survey Questionnaire

"CookLabs: A Web-Based Learning Platform for Cookery NC II students of IETI Marikina"

Respondents Profile

Name (optional): _____ Age:_____

Gender (M/F):_____ Year Level: _____

Directions: Kindly check the appropriate box that corresponds to your indicated answer. Do not leave any blank questions.

Legend:

4- Strongly Agree (SA)	**3 - Agree (A)**	**2 - Disagree (DA)**	**1-Strongly Disagree (SD)**

Part 1: Access to Learning Materials

Questions	SA 4	A 3	DA 2	SD 1
1.1 I can view the modules anytime and anywhere with a stable internet connection.				
1.2 The materials provided are relevant to the Cookery NC II curriculum.				
1.3 I did not experience difficulty locating specific topics or lessons.				
1.4 The content is well-organized and easy to follow.				

Part 2: Progress Tracking

Questions	SA 4	A 3	DA 2	SD 1
2.1 I am able to track which modules I have completed.				
2.2 The progress tracking motivates me to finish lessons.				
2.3 The visual indicators (such as progress bars or checkmarks) help me stay organized.				
2.4 The progress tracking feature is easy to understand and use.				

Part 3: Assessment Tools

Questions	SA 4	A 3	DA 2	SD 1
3.1 The quizzes and activities match the content covered in the lessons.				
3.2 The platform provides instant feedback after I complete an assessment.				
3.3 I find the instructions in each quiz or assessment clear and understandable.				
3.4 I feel confident that the assessment results reflect my actual understanding.				